

Lab 15: Naïve Bayes – Spam Classification

Aim

To implement a **Naïve Bayes classifier** to classify messages as **Spam or Ham (Not Spam)**.

Algorithm

1. Import the required libraries.
2. Create a dataset containing messages and their labels.
3. Convert the text messages into numerical features using CountVectorizer.
4. Split the dataset into training and testing sets.
5. Create a Multinomial Naïve Bayes classifier.
6. Train the model using the training data.
7. Predict whether the test messages are Spam or Ham.
8. Display the predictions and accuracy.

Program

```
# Lab 15: Naïve Bayes – Spam Classification
```

```
from sklearn.feature_extraction.text import CountVectorizer
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.naive_bayes import MultinomialNB
```

```
from sklearn.metrics import accuracy_score
```

```
# Sample dataset
```

```
messages = [
```

```
    "Win a free lottery now",
```

```
    "Hi, how are you?",
```

```
    "Claim your free prize",
```

```
    "Let's meet tomorrow",
```

```
"Congratulations you won money",  
"Can you call me later?",  
"Get free cash now",  
"See you at college"  
]
```

```
labels = [  
    "Spam",  
    "Ham",  
    "Spam",  
    "Ham",  
    "Spam",  
    "Ham",  
    "Spam",  
    "Ham"  
]
```

```
# Convert text into numerical features
```

```
vectorizer = CountVectorizer()
```

```
X = vectorizer.fit_transform(messages)
```

```
# Split the dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, labels, test_size=0.25, random_state=42  
)
```

```
# Create and train the model
```

```
model = MultinomialNB()
```

```
model.fit(X_train, y_train)
```

```
# Make predictions
```

```
y_pred = model.predict(X_test)
```

```
print("Predictions:", y_pred)
```

```
print("Actual Values:", y_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

Predictions: ['Ham' 'Spam']

Actual Values: ['Ham', 'Spam']

Accuracy: 1.0

Result

Thus, the **Multinomial Naïve Bayes classifier** was successfully implemented to classify text messages as **Spam or Ham**.